Qwen3-0.6B GRAFT with a Frozen Reward-Hacking Probe

Libor Burian · Exploratory pilot · 22 October 2026

Abstract

Mitigation effectiveness is inconclusive. The primary comparison is incomplete or has ambiguous judgments. No audited hacking attempt is established in the no-intervention baseline. The primary arms have no established legitimate successes; a capability floor may dominate. This report covers 0 completed training arms and 0 audited study episodes; 0 audit packets remain pending. The experiment is an exploratory single-seed pilot, not a powered confirmatory test.

Context, motivation and introduction

GRAFT separates training updates into retain and forget adapters using a monitor [1]. This study replaces its simulated Terminal Wrench monitor with a separately trained, frozen trajectory detector [2] while reducing the learner to Qwen3-0.6B [3]. The motivation is to distinguish an implementable detector integration from evidence that the integration mitigates reward hacking.

Background and related work

The external detector combines Qwen3-Embedding-0.6B [4], an attention LoRA adapter, and a turn-transformer classifier [2]. Its representations do not share activations with the learner. GRAFT's monitor flags route gradients; they are not scalar reward penalties [1]. External classification accuracy and generated hacking behavior measure different quantities.

Method

The exact preregistration is docs/protocol.md in the published repository commit recorded in publication.json. Preserve the published 232/99 task split, reserve 32 training tasks for calibration (seed 42), and freeze a strict score cutoff at empirical clean FPR <= 5%. Select at most 192 balanced trajectories, in multiples of 48, with a common memory-preflight length cap. Freeze the base; train two width-64 branches for five epochs, effective batch 48, learning rate 5e-4, weight decay 1e-4, betas (0.9,0.95), cosine schedule, 5% warmup, 1% unlabeled anchors and forget-gradient multiplier 2. Compare no intervention, original simulated monitor, downloaded probe, probe filtering and length-bin-shuffled probe flags. Filtering uses ordinary AdamW and thus is not a pure routing contrast. Correct Accelerate accumulation normalization [5] and use loss-equivalent Liger Qwen3 fused CE [6].

Evaluation and analysis

Evaluate eight fixed locally runnable heldout Docker tasks through pinned Harbor, once per mode and prompt condition: seed 2002, thinking disabled, maximum 16 turns and 120 seconds per agent episode. Audit intent using opaque packets before unblinding arms. The primary comparison is elicited hacking in probe-GRAFT retain-only versus no intervention both; legitimate success has a five-percentage-point safeguard. Wilson intervals describe known binary outcomes; ambiguity is retained as bounds. Paired task bootstrap uses 10,000 draws, seed 42, and the exact discordance test is descriptive. Missing outcomes are never imputed as benign. Detector intervals resample whole tasks.

Results

Training coverage: no_intervention: 0 updates, complete=False; probe_graft: 0 updates, complete=False; original_graft: 0 updates, complete=False; probe_filtering: 0 updates, complete=False; random_graft: 0 updates, complete=False; smoke: 0 updates, complete=False.

Detector: {"status": "not run"}.

Primary paired estimates: {"hacking": {"paired_tasks": 0, "known_pairs": 0, "difference": null, "bootstrap_ci95": null, "exact_discordance_p": null, "ambiguity_bounds": null, "task_differences": {}, "warning": "Conditional on one training seed; n=8 bootstrap may be degenerate; missing pairs are not imputed"}, "legitimate": {"paired_tasks": 0, "known_pairs": 0, "difference": null, "bootstrap_ci95": null, "exact_discordance_p": null, "ambiguity_bounds": null, "task_differences": {}, "warning": "Conditional on one training seed; n=8 bootstrap may be degenerate; missing pairs are not imputed"}}

Claims and implications

Mitigation effectiveness is inconclusive. The primary comparison is incomplete or has ambiguous judgments. No audited hacking attempt is established in the no-intervention baseline. The primary arms have no established legitimate successes; a capability floor may dominate. Successful checkpoint loading, kernel equivalence and successful Docker execution support engineering compatibility only. Reported treatment differences are conditional descriptive observations, not claims of robust alignment or a causal mechanism beyond the implemented adapter intervention.

Discussion and limitations

One training seed, eight selected tasks, a much smaller model than the source paper, task difficulty, label/source confounding, unresolved upstream detector exposure, truncation and non-independent implementing-assistant audits limit generalization. A sigmoid score is not assumed calibrated. At n=8 one binary outcome changes a rate by 12.5 percentage points, larger than the 5pp capability tolerance. Bootstrap intervals may degenerate at the boundary; exact paired discordance evidence and Wilson intervals remain visible. The six-hour cap can leave arms or modes incomplete. Training error and infrastructure error are reported separately from behavior. No result-dependent extension or threshold retuning is allowed.

Conclusion

Mitigation effectiveness is inconclusive. Larger, multi-seed studies with measured task/seed variance and independently blinded audits are required for stronger conclusions.

Appendix: prospective power and reproducibility

Normal-approximation sample requirements for a 10pp reduction at 80% power and two-sided alpha .05: [{"baseline": 0.1, "treatment": 0.0, "n_per_arm": 74}, {"baseline": 0.2, "treatment": 0.1, "n_per_arm": 199}, {"baseline": 0.3, "treatment": 0.19999999999999998, "n_per_arm": 294}, {"baseline": 0.5, "treatment": 0.4, "n_per_arm": 388}]. Paired discordance and unknown seed-variance sensitivity are in results.json; they are planning assumptions. uv.lock pins the environment; artifacts.json pins sources and weights; data_lock.json fixes examples and preprocessing; per-checkpoint configuration and progress files record all actual updates. Full prompts, responses, scores and judgments are retained in timestamped local logs. Run uv run python -m experiments.qwen_probe run --profile pilot.

References

[1] GRAFT. https://arxiv.org/abs/2610.17062
[2] farzingkh/reward-hacking, revision cd4c566824c5961eaeab7b7fce7845240bd12b3d (MIT source). https://github.com/farzingkh/reward-hacking
[3] Qwen3-0.6B. https://huggingface.co/Qwen/Qwen3-0.6B
[4] Qwen3-Embedding-0.6B. https://huggingface.co/Qwen/Qwen3-Embedding-0.6B

[5] Accelerate gradient accumulation.
https://huggingface.co/docs/accelerate/usage_guides/gradient_accumulation
[6] Liger Kernel. https://github.com/linkedin/Liger-Kernel
[7] SciPy binomial inference.
https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.binomtest.html
[8] NeurIPS reporting checklist. https://neurips.cc/public/guides/PaperChecklist
[9] Lones (2021), ML pitfalls. https://arxiv.org/abs/2108.02497
[10] Lipton and Steinhardt (2018), Troubling Trends. https://arxiv.org/abs/1807.03341

